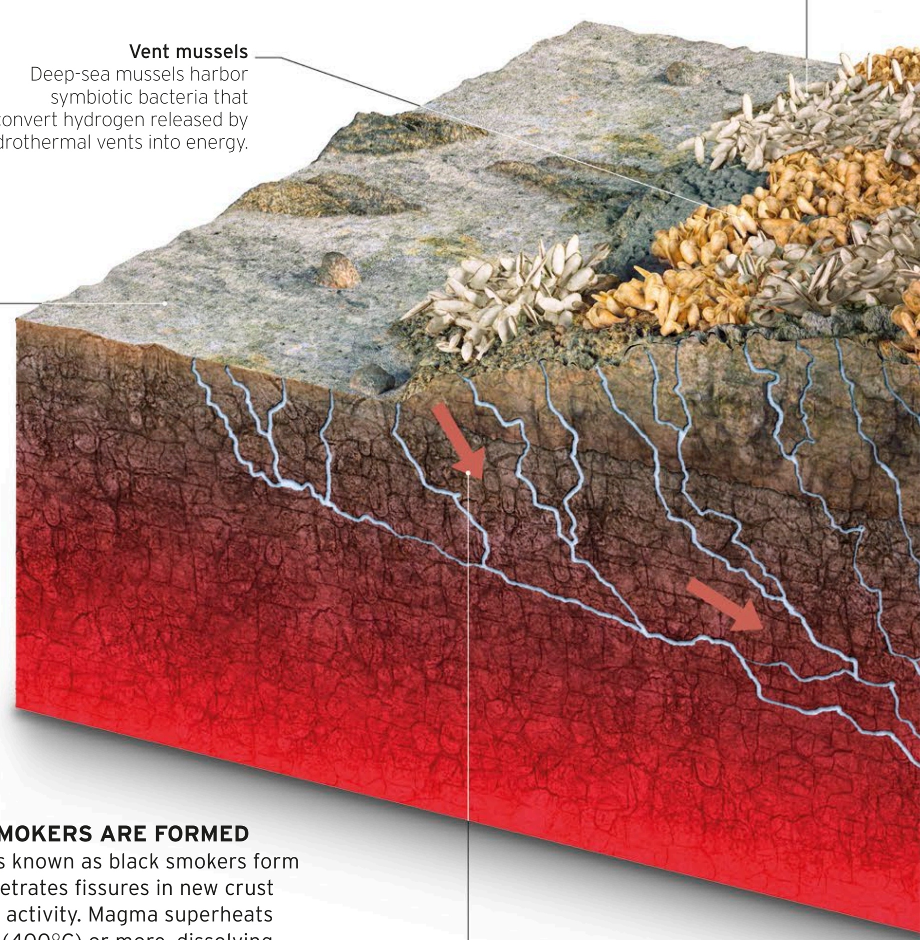
The East Pacific Rise

A volcanic ridge deep on the ocean floor, where life thrives in the harshest of conditions

The East Pacific Rise is part of a volcanic mid-oceanic ridge system that extends along the floor of the southeastern Pacific Ocean. It lies at a depth of about 1½ miles (2.4 km) and rises about 5,900–8,900 ft (1,800–2,700 m) above the sea floor.

Oceanic ridges occur where tectonic plates move apart (see pp.312–13). The process, called rifting, creates fissures along the seabed, releasing lava that forms a ridge as it cools. New basaltic lava is added to either side of its crest as the plates spread apart. Compared to the Mid-Atlantic Ridge, the East Pacific Rise spreads relatively quickly and lacks a pronounced central rift valley. The rifting process produces geyser-like hydrothermal vents that discharge heated, mineral-laden seawater. Despite toxic sulfides, high temperatures, and enormous underwater pressures, the vents support an amazing community of creatures, from single-celled organisms to giant tubeworms.

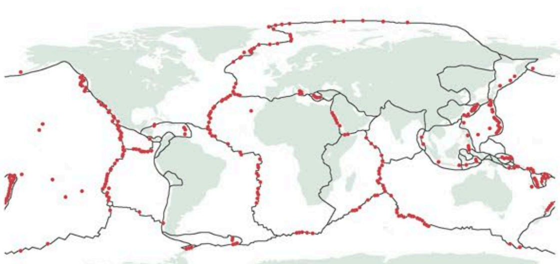
◁ **STRANGE FLOWER**
The ocean dandelion is a type of deep-sea siphonophore, a group of animals that includes jellyfish and the Portuguese man o' war. It is not one, but a collection of individuals that share tissues.



▶ **HOW BLACK SMOKERS ARE FORMED**
Hydrothermal vents known as black smokers form when seawater penetrates fissures in new crust created by volcanic activity. Magma superheats the water to 750°F (400°C) or more, dissolving material from surrounding rocks. This superhot vent fluid rises back through openings in the seabed. As they cool, the dissolved minerals precipitate into solid form, creating mineral- and metal-rich chimneylike vents.

THE WORLD'S VENT FIELDS

The first hydrothermal vents were discovered in 1977 by an expedition exploring a spreading ridge near the Galápagos Islands. Vent fields have since been recorded in all of the oceans. They are formed in volcanically active areas, which is why so many are found along diverging plate boundaries that coincide with mid-oceanic ridges. Others are created by hotspots (see p.318) or where tectonic plates converge and one pushes beneath (subducts) the other.



More than **500 species** have been found living at **hydrothermal vents**